

DNS Security

Project Overview

This project was initiated to assess and enhance the security posture of the Domain Name System (DNS) within the network environment. The primary objective is to protect DNS infrastructure from cyberattacks and ensure fast, reliable, and secure domain resolution services.

The project focuses on the following goals:

- Identify and address DNS-related threats, such as DNS spoofing, tunneling, hijacking, and cache poisoning.
- Implement layered DNS security controls, including DNSSEC, DNS filtering, encrypted DNS (DoH/DoT), and DNS firewalls.
- Harden DNS infrastructure by separating authoritative and recursive servers, locking DNS cache, and securing zone transfers.
- Enable comprehensive logging and monitoring of DNS queries and server behavior to detect malicious activity early.
- Ensure DNS redundancy and high availability to prevent service outages and support network resilience.
- Establish clear DNS security policies and access controls to maintain operational integrity and reduce attack surfaces.

The project outcome delivers a robust and secure DNS infrastructure that protects against modern DNS-based threats, ensures reliable name resolution, and provides a strong foundation for overall network security.

Description

DNS security is the practice of protecting DNS infrastructure from cyber attacks in order to keep it performing quickly and reliably. DNS security solutions create an extra layer of security between a user and the internet. An effective DNS security strategy incorporates a number of overlapping defenses, including establishing redundant DNS servers, applying security protocols like DNSSEC, and requiring rigorous DNS logging.

Recommendations

A secure and maintained DNS infrastructure is essential for uninterrupted network operations. Following these tips ensures the system is secure and has optimal performance.

- Ensure DNS Redundancy and High Availability
- Restrict Access to DNS Servers Information
- Use Internal DNS Servers for Private Queries
- Use Local or Closest DNS Servers
- Enable DNS Logging and Monitoring
- Lock DNS Cache
- Filter Requests
- Validate DNS Data Integrity with DNSSEC
- Implement Rate Limiting
- Update DNS Software
- Separate Authoritative and Recursive DNS Servers
- Implement DDoS Protection
- Perform Periodic Testing
- Create Backups
- Secure Zone Transfers

References

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Details

Domain Name System (DNS) is a method for cataloging the IP addresses of servers and their associated web page URL addresses. Since DNS is responsible for enabling all internet activity, keeping an eye on DNS requests and the IP addresses they lead to can help keep your network secure

If a cybercriminal infiltrates a DNS system, they can send users to fake or malicious sites. They can also steal data, hijack websites, or inundate servers with requests, shutting them down eventually. DNS security is designed to prevent these kinds of attacks.

Attackers have found a number of ways to target and exploit DNS servers. Here are some of the most common DNS attacks:

- *DNS Spoofing/Cache Poisoning*: Fake DNS data is injected into a resolver's cache, sending users to malicious websites.
- *DNS Tunneling*: Attackers hide data inside DNS queries/responses to bypass firewalls.
- *DNS Hijacking*: Redirects DNS queries to malicious servers by altering DNS settings, often via malware.
- *NXDOAMIN Attack*: Floods DNS with requests for nonexistent domains to exhaust resources and cause denial-of-service
- *Phantom Domain Attack*: Floods a resolver with slow or non-responsive domain requests, tying up system resources.
- *Botnet-Based CPE Attack*: Uses hijacked home routers/devices to launch large-scale DNS attacks like

subdomain floods.

- *Random Subdomain Attack:* Sends queries for many fake subdomains to overwhelm the authoritative DNS server.

To prevent DNS leaks and attacks, secure your DNS servers and block suspicious queries. DNS-layer protection stops malware communication and prevents hackers from hijacking your DNS. By compiling a list of risky websites and filtering out undesired content, DNS security solutions create an extra layer of security between a user and the internet. As a result, your Domain Name System (DNS) will no longer be exposed to dangers or potentially harmful assaults.

DNSSEC

DNSSEC is a feature of the Domain Name System (DNS) that uses cryptographic authentication to verify that the DNS records returned in a DNS query come from an authoritative name server and are not altered en route. DNSSEC (short for Domain Name System security extensions) is used to extend DNS protocol and address vulnerabilities in the DNS that leave the system susceptible to various cyberattacks, such as DNS spoofing, DNS cache poisoning, man-in-the-middle attacks, and other unauthorized modifications to DNS data.

DNS filtering:

This technology screens and blocks DNS requests for unwanted or malicious websites to reduce the risk of malware infections and data exfiltration.

DNS Firewalls:

A DNS firewall works by filtering the traffic that moves along DNS endpoints. This filtration process checks all the traffic using specific rules and policies. If the firewall finds that the traffic violates one of these policies or rules, the web request gets blocked.

Secure DNS servers:

DNS servers that support DNS over TLS (DoT) or DNS over HTTPS (DoH) enable DNS traffic to be encrypted, preventing attackers from manipulating or eavesdropping on DNS traffic.

Affected Locations

- Client devices
- Local DNS Resolver
- Authoritative DNS Servers
- Recursive Resolvers
- DNS Caches